PC Command Prompt Reference

Networking commands for testing and troubleshooting from an end device

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These run from a PC's Command Prompt (in Packet Tracer: Desktop → Command Prompt; on a real Windows machine: search "cmd"). This is the toolkit you use to prove a network works — and to find out where it's broken when it doesn't. Commands marked **PT** work in Packet Tracer; the rest are real-Windows only but worth knowing for the job.

1. Connectivity Testing

Command	*What it does / when to use it*
`ping 192.168.20.11` **[PT]**	Sends test packets to an address and reports replies. The core "can I reach it?" test.
`ping 192.168.10.1` **[PT]**	Ping your own gateway first — if this fails, the problem is local, not remote.
`ping -n 10 192.168.20.11`	Sends exactly 10 packets instead of the default 4 (Windows).
`ping -t 192.168.20.11`	Pings continuously until you press Ctrl+C — useful for watching an intermittent connection.
`tracert 192.168.99.13` **[PT]**	Shows every router hop between you and the destination — pinpoints where traffic stops.
`pathping 192.168.20.11`	Combines ping and tracert, adding packet-loss stats per hop (Windows).

Reading ping results: `Reply from...` = success. `Request timed out` = no response (device down, firewall, or wrong route). `Destination host unreachable` = no route exists at all. A single timeout on the *first* ping is usually just ARP resolution, not a fault.

TTL tells you hop count: a reply with TTL=128 stayed on your own subnet; TTL=127 crossed one router; TTL=126 crossed two. Handy for confirming inter-VLAN routing is actually happening.

2. Viewing & Managing IP Configuration

Command	*What it does / when to use it*
`ipconfig` **[PT]**	Shows your IP address, subnet mask, and default gateway. First command to run on any "no internet" complaint.
`ipconfig /all` **[PT]**	Adds MAC address, DHCP server, DNS servers, and lease times — the fuller picture.
`ipconfig /release` **[PT]**	Gives up the current DHCP lease (address becomes 0.0.0.0).
`ipconfig /renew` **[PT]**	Requests a fresh DHCP lease. Release + renew is the classic fix for a bad address.

Command	*What it does / when to use it*
`ipconfig /flushdns`	Clears the local DNS cache — use when a site resolves to a stale/wrong address (Windows).
`ipconfig /displaydns`	Shows what's currently in the DNS cache (Windows).

3. ARP & MAC Address Tools

ARP maps IP addresses to MAC addresses on the local network. Useful for spotting duplicate IPs and confirming who's actually answering.

Command	*What it does / when to use it*
`arp -a` **`[PT]`**	Lists cached IP-to-MAC mappings this PC has learned.
`arp -d`	Clears the ARP cache — forces fresh resolution on the next ping (Windows).
`getmac`	Shows this machine's own MAC address(es) (Windows).

4. DNS Lookup

Command	*What it does / when to use it*
`nslookup google.com` **`[PT]`**	Asks DNS to resolve a name to an IP. If ping-by-IP works but ping-by-name doesn't, it's a DNS issue.
`nslookup google.com 8.8.8.8`	Queries a specific DNS server instead of the default — tests whether your configured DNS is the problem.

5. Connections, Ports & Routing Table

Command	*What it does / when to use it*
`netstat -an`	Lists all active connections and listening ports, numerically (Windows).
`netstat -b`	Adds which program owns each connection — needs admin rights (Windows).
`route print`	Displays this machine's routing table — where it sends traffic for each destination (Windows).
`netsh interface show interface`	Lists network adapters and their status (Windows).
`netsh wlan show profiles`	Lists saved Wi-Fi networks on this machine (Windows).

6. Remote Access & File Transfer

Command	*What it does / when to use it*
`telnet 192.168.10.1` **`[PT]`**	Opens a Telnet session to a device (e.g. your router). Unencrypted — lab use only.
`ssh -l admin 192.168.10.1` **`[PT]`**	Opens an encrypted SSH session. Preferred over Telnet in any real environment.
`hostname`	Shows this machine's name (Windows).

7. Windows Commands Worth Knowing for IT Roles

Not Packet Tracer commands, but common in day-to-day support and admin work — useful for the Office Admin / IT side of the project.

Command	*What it does / when to use it*
`systeminfo`	Full system report: OS version, install date, RAM, domain membership, patches.
`tasklist`	Lists all running processes (command-line Task Manager).
`taskkill /IM notepad.exe /F`	Force-closes a process by name.
`sfc /scannow`	Scans for and repairs corrupted Windows system files (run as admin).
`chkdsk C: /f`	Checks a drive for errors and fixes them (run as admin).
`gpupdate /force`	Forces a Group Policy refresh — use after making AD/GPO changes.
`gpresult /r`	Shows which Group Policies currently apply to this user and machine.
`whoami`	Shows the currently logged-in user account, including domain.
`net user`	Lists local user accounts on the machine.
`shutdown /r /t 0`	Restarts immediately (/s instead of /r shuts down).

8. A Practical Troubleshooting Order

When someone reports "the network is down," work outward from the machine. Each step narrows where the fault can be:

Step	*Command*	*What a failure tells you*
1. Check own config	`ipconfig /all`	No IP or a 169.254.x.x address = DHCP isn't reaching this device
2. Ping own gateway	`ping 192.168.10.1`	Fails = local link, VLAN, or cabling problem

Step	*Command*	*What a failure tells you*
3. Ping across VLANs	`ping 192.168.20.11`	Fails = routing or ACL problem on the router
4. Trace the path	`tracert 192.168.99.13`	Shows exactly which hop traffic dies at
5. Test name resolution	`nslookup google.com`	Fails while IPs work = DNS misconfiguration

This order is worth memorising — being able to describe it in an interview shows structured thinking rather than guesswork.

Reference sheet prepared for the Small Office IT Infrastructure Lab project. Pair with the Cisco switch and router command references.